

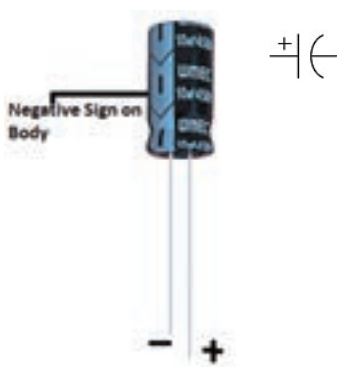
2.2.3 Capacitor

A capacitor is a device used to store electric charge. Their ability to store charge is measured in 'Farads'. The more the 'Farad' rating of the capacitor, the more charge it can store. But, Farad is a very large unit, so we commonly use smaller unit such as 'microfarads' to denote the capacitance of a capacitor.

Polar Capacitors

These capacitors have fixed positive and negative terminals. These capacitors have to be connected in the correct way in the circuit. If you connect it in reverse polarity (connecting positive terminal where negative is to be connected and vice versa), an explosion may occur and you may spoil quite a few electronic components of your circuit.

The most widely available polar capacitors are electrolytic capacitors. You can identify the polarity of the terminals of an electrolytic capacitor by comparing their lengths. The longer terminal is always positive. Also, on the body of the capacitor, there is a negative sign mentioned above the negative terminal.



Electrolytic capacitors are polarised

Non Polar Capacitors

These capacitors do not have any polarity and can be connected in either way in a circuit. Most commonly used non polar capacitors are mica and ceramic capacitors. Also, because both the terminals of the non-polar capacitor are identical, the symbol for them is slightly different than that for polar capacitors.

Identifying capacitance values

A polar capacitor has both the capacitance value and the voltage rating clearly written on its body. However, for mica capacitors, they are written on the body in encoded form, generally a three digit number. If the number written on body is 474, then the value is 47×10^4 . That is, the first two digits are written as they are, and the last digit is considered to be power of 10. Please note that the capacitances when decoded, give value in pico farads (pf). So 474 is equal to 47×10^4 pf and 342 is equal to 34×10^2 pf.